

Persistent Industrial Thermal Source Detection

A leakage-controlled FIRMS, Sentinel-2, and WorldCover ranking system for prioritizing persistent industrial heat sources.

FROZEN RANKING MODEL	guarded_cv_tabular protocol 12b v1
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0.9293 MACRO COUNTRY PR-AUC	+0.0239 PR-AUC VS COMPACT	0.8166 MACRO COUNTRY F1*	5 / 6 COUNTRIES IMPROVED
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Decision. Promote the guarded visual residual over the compact branch for ranking. All 12 preregistered confirmation gates passed across three fresh seeds on the same six-country panel.

Boundary. The deployment threshold is not frozen. India was never loaded, and the reported F1 uses held-country thresholds rather than one production cutoff.

Freeze ID SIH-NB12B-2026-09-04

Source model commit 288de63

01 Problem, data, and supervision

The task is source ranking under incomplete labels, not row-level fire classification.

78 FIRMS CSV FILES	19.34M DETECTIONS AUDITED	7 COUNTRIES, 2019-2024	1 km SOURCE GRID
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From detections to physical sources

Stage	Observed input	Design choice	Why it matters
Audit	19,342,072 detections	Validate dates, schema, duplicates, FRP	Clean inputs before aggregation
Clustering	Libya: 427,517 rows	1 km metric grid -> 4,824 sources	Avoid training on near-duplicate events
Splitting	One flare fragments into about 4 sources	Group by 10 km block	Keep fragments out of opposing folds
Common window	Unequal country coverage	Use 2022-2024	Remove time coverage as a country proxy

Positive-unlabeled framing

EOG identifies gas flares, not every industrial thermal source. A kiln, cement plant, or steel facility may be truly industrial while unmatched by EOG. Matches are treated as positives; unmatched sources remain unlabeled rather than verified negatives.

Final confirmatory panel

Country	Sources	EOG positives	Country	Sources	EOG positives
Algeria	48	15	Angola	50	6
Indonesia	48	17	Iraq	48	18
Libya	51	15	Nigeria	49	16
Total	294	87	India	Not loaded	Not loaded

This is an enriched QA panel, not deployment prevalence. The 1,342 earlier labels refer to active flare inventory sites across the four originally labelled training countries.

02 How the model evolved

Each revision addressed a measured transfer or leakage risk rather than adding complexity for its own sake.

Phase	Approach	Observation	Decision
FIRMS baselines	Isolation Forest, rules, LightGBM	Early Libya + Algeria LightGBM reached F1 0.691, PR-AUC 0.740	Keep tree model as the base, but stop treating regional CV as final
Transfer correction	2022-2024 common window, PU framing, country LOCO	Country imbalance and background shift explained optimistic validation	Use held-country evaluation and country-safe preprocessing
Compact multimodal	FIRMS plus compact Sentinel-2 and WorldCover summaries	NB9 macro country PR-AUC about 0.9059 on the 294-source panel	Protect this as the compact baseline
Guarded CV	Frozen dense embeddings as an uncertainty-weighted residual	NB12 discovery: 0.9283 vs 0.9082 PR-AUC	Confirm only this challenger with fresh fixed seeds
NB12b confirmation	Seeds 272, 273, 274; 12 fixed gates	Guarded reached 0.9293 vs 0.9054 PR-AUC	Freeze guarded ranking model

Historical figures use different cohorts and protocols. They explain design evolution and must not be compared as one leaderboard.

Leakage controls retained in the final design

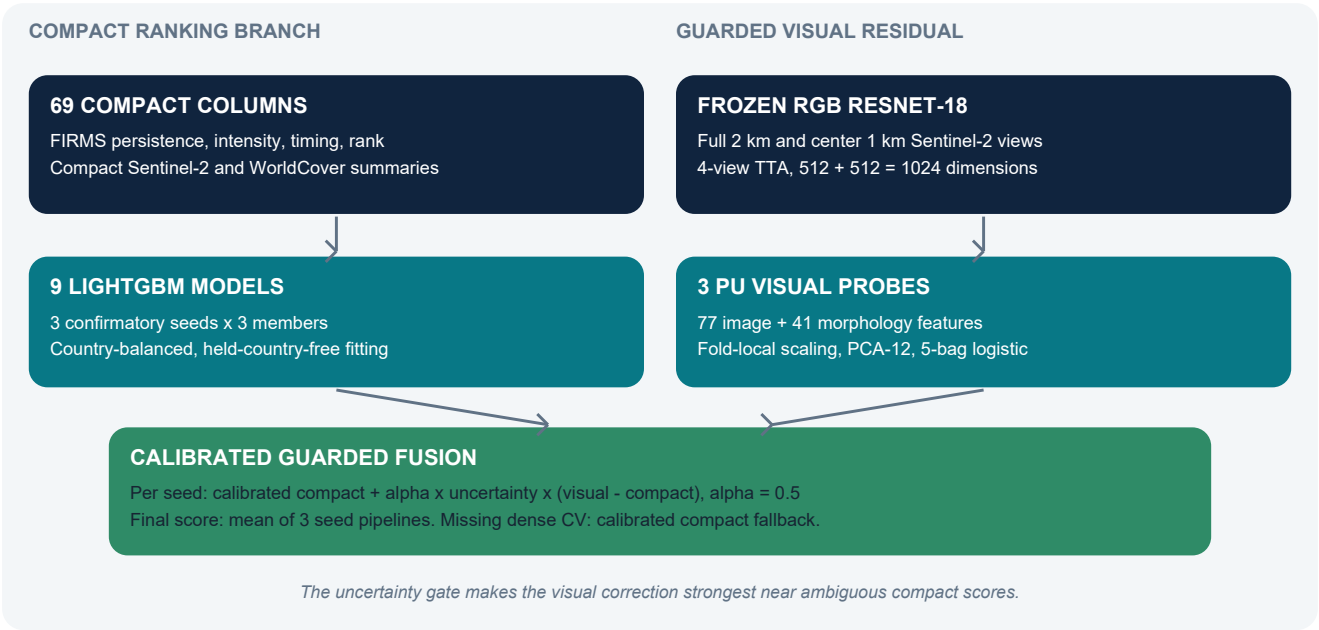
- Exclude latitude, longitude, country, NASA type, and EOG distance from learned features.
- Remove raw NOAA-20 cross-instrument signals when availability would identify Angola or Indonesia.
- Fit normalization, PCA, visual probes, LightGBM, fusion weight, and validation thresholds without the held country.
- Keep India sealed until the foreign ranking model and evaluation policy are frozen.

Why a residual instead of end-to-end CV

Only 294 QA-passing sources were available for the dense visual branch. A frozen encoder plus regularized PU probes limits overfit, while the compact branch remains the stable prior. Visual evidence can correct ambiguous ranks without replacing the trusted base score.

03 Frozen model architecture

The selected model is a three-seed ensemble of calibrated compact rankings plus a bounded visual correction.



Exact fusion rule

$$\text{fused} = \text{tabular} + \alpha * 4 * \text{tabular} * (1 - \text{tabular}) * (\text{visual} - \text{tabular})$$

Alpha is 0.5 for seeds 272, 273, and 274. The final score is the mean of the three seed pipelines.

Frozen components

Component	Frozen specification	Count
Compact schema	69 FIRMS plus compact Sentinel-2 and WorldCover columns	1 schema
Compact learners	LightGBM, three members for each confirmatory seed	9 models
Visual encoder	Pinned Sentinel-2 RGB MoCo ResNet-18 checkpoint, no fine-tuning	1 checkpoint
Visual representation	2 km full view + 1 km center view, four-view TTA	1024 values
Visual context	77 image and land-cover features + 41 morphology features	118 values
Visual probes	Fold-local L2 normalization, scaling, PCA-12, five PU bags	3 models
Fallback	Mean of three seed-specific calibrated compact pipelines	Deterministic

The missing-image fallback is calibrated compact scoring. It has not been validated as a fully imagery-free India deployment mode because the compact schema itself includes compact imagery summaries.

04 Validation and integrity design

The confirmation was fixed before the run, used fresh seeds, and kept every held country outside model selection.

Layer	Rule	Evidence produced
Outer evaluation	Leave one country out across six foreign countries	Held-country predictions for every source
Spatial grouping	10 km block IDs	Fragments of one physical site remain grouped
Inner fitting	All preprocessing, models, alpha, and F1 threshold exclude outer country	Country-safe fold records
Fresh seeds	272, 273, 274 fixed; discovery seed 271 excluded	Three independent algorithmic reruns
Uncertainty	5,000 positive-stratified 10 km block repeats, seed 12012	Paired PR-AUC gain interval
Holdout	India never loaded	No holdout-driven selection
Integrity	Hashes for cohort, caches, checkpoint, code, predictions, and models	33 declared artifacts verified

Preregistered promotion gates

Gate	Required	Observed	Pass
Integrity valid	True	True	Yes
Macro PR-AUC gain	$\geq +0.010$	+0.023932	Yes
Countries improved	≥ 4	5	Yes
Worst country PR-AUC delta	≥ -0.020	-0.005054	Yes
Indonesia PR-AUC delta	≥ -0.020	-0.005054	Yes
Macro F1 delta	≥ -0.010	+0.049744	Yes
Bootstrap lower 95%	≥ -0.010	-0.000632	Yes
Macro recall at top 20% delta	≥ -0.020	-0.011111	Yes
Worst positive-count loss	≥ -1	-1	Yes
Every seed PR-AUC gain	> 0	3 of 3	Yes
Median seed gain	$\geq +0.010$	+0.022842	Yes
Nonzero seed alphas	≥ 2	3	Yes

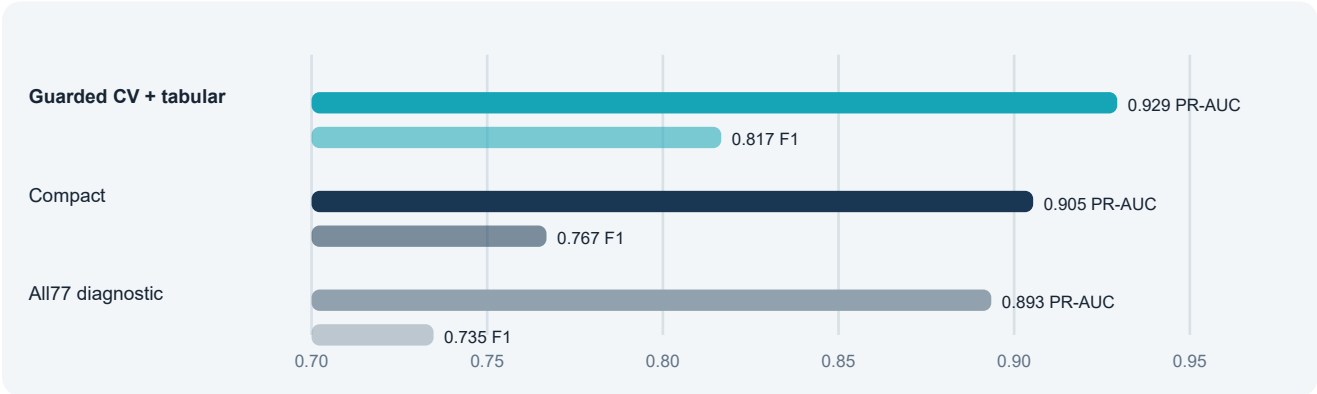
All 77 and the nested selector remained diagnostic only and could not change the promotion decision.

05 Headline confirmatory results

Guarded CV improves ranking and thresholded F1 on the enriched foreign panel, with stable gains across fresh seeds.

Branch	Macro F1*	Macro PR-AUC	Macro ROC-AUC	Worst-country PR-AUC
Guarded CV + tabular	0.816620	0.929336	0.948094	0.840037
Compact	0.766876	0.905404	0.919853	0.825015
All77 diagnostic	0.734732	0.893485	0.922493	0.848884

*Macro F1 uses country-specific median fold-local LOCO thresholds. It is not performance at one frozen deployment threshold.



Observed gain

Guarded vs compact: +0.023932 macro PR-AUC, +0.049744 macro F1, and +0.028241 macro ROC-AUC. Five of six countries improved in PR-AUC.

Fresh-seed stability

Seed	Compact PR-AUC	Guarded PR-AUC	PR-AUC gain	F1 gain	Alpha
272	0.906547	0.929960	+0.023413	+0.051996	0.5
273	0.905569	0.928411	+0.022842	+0.047596	0.5
274	0.908584	0.930390	+0.021806	+0.053570	0.5

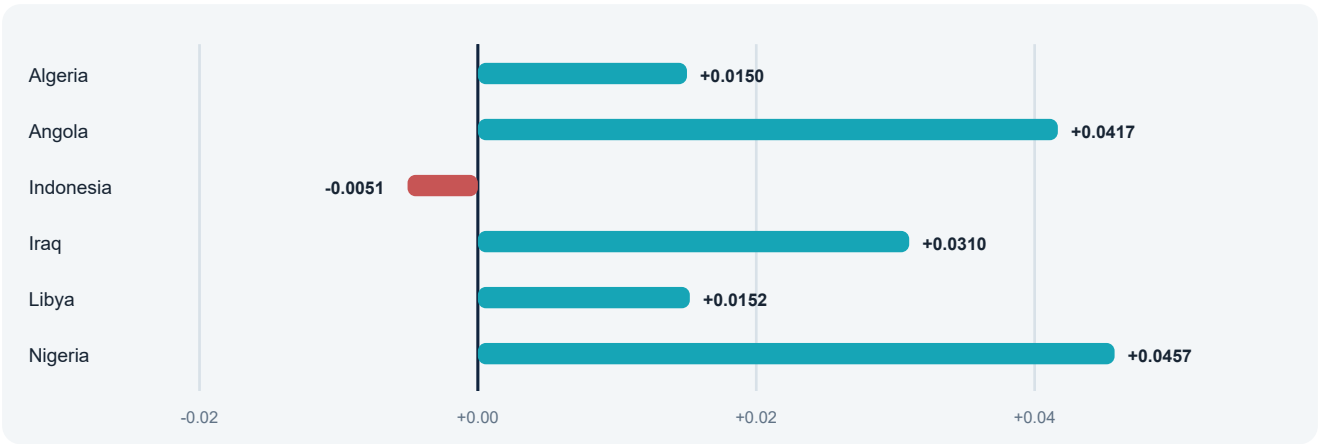
All three fresh seeds gained 2.18 to 2.34 PR-AUC points on the same panel. This tests algorithmic seed variance, not new-country or new-sample variance.

06 Country-level behavior

The visual residual helps most in Nigeria and Angola. Indonesia is the only PR-AUC decline and remains within the fixed guard.

Country	Compact PR-AUC	Guarded PR-AUC	Delta	Compact F1*	Guarded F1*
Algeria	0.825015	0.840037	+0.015022	0.666667	0.714286
Angola	0.958333	1.000000	+0.041667	0.909091	0.923077
Indonesia	0.955917	0.950863	-0.005054	0.692308	0.857143
Iraq	0.877719	0.908707	+0.030988	0.848485	0.875000
Libya	0.909849	0.925075	+0.015226	0.764706	0.789474
Nigeria	0.905589	0.951333	+0.045744	0.720000	0.740741

*F1 uses country-specific median fold-local LOCO thresholds. Angola PR-AUC 1.000 is based on six EOG positives and should not be read as population perfection.



PR-AUC change, guarded minus compact. Indonesia falls 0.51 points, inside the preregistered -2.0 point guard.

Interpretation

The country pattern supports a bounded visual correction rather than a broad replacement of the compact model. Improvement is distributed, but the cohort is too small to infer industrial population precision or India transfer.

07 Uncertainty and review-budget behavior

Ranking promotion passed the fixed uncertainty guard, while the top 20% operating slice retained one fewer labelled positive.

Statistic	Result	Correct reading
Paired block-bootstrap mean	+0.023764 PR-AUC	Positive average ranking gain
95% interval	[-0.000632, +0.051502]	Crosses zero
Prespecified lower guard	>= -0.010	Passed
Bootstrap design	5,000 repeats	Positive-stratified within-country 10 km blocks

Statistical claim boundary

The interval crosses zero. The result supports promotion under the prespecified non-inferiority lower guard, but it is not conventional positive statistical significance.

Top 20% review budget

Country	Reviewed	Compact found	Guarded found	Change
Algeria	10	8	8	0
Angola	10	6	6	0
Indonesia	10	10	10	0
Iraq	10	10	10	0
Libya	11	11	10	-1
Nigeria	10	10	10	0
Total	61	55	54	-1

Guarded recovered 54 of 87 EOG matches versus 55 for compact. Macro recall at the top 20% was 0.661465 vs 0.672576. Ranking uplift did not improve this operating slice.

Recommended India execution pattern



Threshold is intentionally open

Do not deploy the pilot threshold 0.616443. Calibrate a review budget or operating cutoff using foreign out-of-fold scores only, then keep it fixed before opening India.

08 Freeze contract, artifacts, and next step

The frozen package is reproducible and auditable. Remaining work is a sealed evaluation step, not another foreign tuning cycle.

Frozen item	Value
Freeze ID	SIH-NB12B-2026-09-04
Source model commit	288de63
Protocol	12b-guarded-confirmatory-ensemble-v1
Selected branch	guarded_cv_tabular
Seeds and alpha	272, 273, 274; alpha 0.5 for each
Artifacts	9 compact models, 3 visual models, calibration reference, schema, caches
Missing dense-CV fallback	Mean of 3 calibrated compact pipelines
Threshold	Not frozen; pilot threshold is not deployment calibrated
Change policy	Any material foreign-data retuning requires a new protocol and freeze ID

Integrity anchors

Artifact	SHA256
Stable panel cohort	7c44d70f3505d65d92bba4abbd2c2e9bebf3eef6a1eb9a432880612ffef3a2f0
Dense embedding cache	44b78d9137b0cebc3f07d6aba2322a0139d402f04c1329fbac84ecc43a74f00b
Morphology cache	b873bf53eaf9161f9d4f775bc84af9670a7d7d52124988a921347894e0127719
ResNet-18 checkpoint	e3a335e38d1d189ad3b0eba4be4004a9c52c5e846317b6737ac9f0fac57e1ac8
Ensemble predictions	6386efa6739e0710182e311cbb36314891afe89eeaad79723741a35132e982b8

NB12b declared 33 artifacts; every file and hash matched. The CPU confirmation completed in 3.14 minutes because dense embeddings were already cached. This was not end-to-end CPU image encoding.

Limits that remain

- The same enriched 294-source panel was reused; only 294 of the intended image batch passed into the final QA cohort.
- Three seeds measure algorithmic variance, not country-sampling or cohort-sampling variance.
- EOG labels gas flares only, so precision and F1 do not measure every industrial thermal source.
- India imagery and labels were not loaded, and no India score exists yet.
- The deployment threshold or fixed review budget remains to be calibrated.

One controlled next run

1. Calibrate the operating policy from foreign out-of-fold scores only. 2. Acquire the matching India compact and dense imagery inputs. 3. Lock code, hashes, threshold or budget, and output schema. 4. Open India once, report the result without retuning, and separate label coverage from model error.

Final status

Ranking model frozen and promoted: guarded_cv_tabular.
Operating threshold: pending deployment-equivalent calibration.
Independent holdout: India remains sealed.